Large Language Models - Exam

Politecnico di Torino

11 Febbraio 2025

Tempo impiegato: 1 ora 15 min. **Valutazione:** 10,38 / 18,00

Domanda 1 *(2 points (no penalty for a wrong answer))*

What is the likely output of the following code snippet?

```python
from transformers import AutoModel, AutoTokenizer

model_name = "bert-base-uncased"
model = AutoModel.from_pretrained(model_name)
tokenizer = AutoTokenizer.from_pretrained(model_name)

prompt = "The result of the operation 2+2 is "
tokens = tokenizer(prompt, return_tensors="pt")

result = model.generate(**tokens)
print(tokenizer.batch_decode(result))

```

Provide a brief explanation (max 50 words). If an error occurs, indicate the line number and briefly explain why.

Domanda 2 *(1 point (-25% penalty for each wrong answer))*

What is the purpose of reflection in an agentic system?

a. To eliminate the need for structured workflows

b. To generate human-like responses using predefined scripts

c. To store knowledge permanently in short-term memory

d. To execute tasks without modifying strategies

e. To evaluate past decisions and improve future performance

Domanda 3 *(1 point (-25% penalty for a wrong answer))*

What is the main difference between encoder-decoder cross-attention and encoder self-attention?

a. In encoder-decoder cross-attention, the decoder attends only to the current token in the input sequence, while in encoder self-attention, the encoder attends to all tokens in the input

b. Encoder self-attention attends to the encoder's output, while encoder-decoder cross-attention attends to the decoder's previous tokens

c. In encoder self-attention, the decoder attends to the input sequence, whereas in encoder-decoder cross-attention, the encoder attends to the output sequence

d. Encoder self-attention is used in the decoder, and encoder-decoder cross-attention is used in the encoder

e. None of the other options is correct

f. Encoder-decoder cross-attention attends only to the previous tokens in the output sequence, while encoder self-attention attends only to tokens in the input sequence

g. Encoder self-attention only uses information from the input sequence, while encoder-decoder cross-attention allows the decoder to attend to the encoder's output

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Domanda 4 *(1.5 points (no penalty for a wrong answer))*

You are given the following sequence of characters:

aceaceaaaaacb

Initially, each character is a separate token. Apply Byte-Pair Encoding to produce 3 additional tokens. What are the generated tokens?

Write each token, in the correct order, in the following slots. Separate the merged tokens with a comma (and no spaces).

Examples:

- if the merged tokens are aa and b, write: aa,b
- if the merged tokens are x and y, write: x,y

Domanda 5 *(1 point (-25% penalty for a wrong answer))*

Which of the following best describes how BERTScore evaluates text similarity?

a. It measures sequence similarity using n-gram co-occurrence weighted by IDF scores

b. It uses a fine-tuned BERT model to classify whether the reference and candidate texts are semantically related or not

c. It predicts a similarity score by treating the evaluation task as a masked language modeling problem, where BERT predicts the score as the next token

d. It assigns similarity scores based on syntactic parse trees rather than token embeddings

e. It uses cosine similarity on static word embeddings trained with Word2Vec or GloVe

f. It applies a bag-of-words model to compare token frequency distributions between reference and candidate texts

g. It computes token-level similarity by aligning contextualized embeddings from a pre-trained transformer model and aggregating scores

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Domanda 6 *(2 points (no penalty for a wrong answer))*

Apply absmax quantization to the following values x_1, x_2, x_3, x_4, to map them to int8.

$$x_1 = -5, \quad x_2 = 4, \quad x_3 = -1, \quad x_4 = 0$$

Write the quantized versions.

Domanda 7 *(1 point (-25% penalty for each wrong answer))*

Which of the following are benefits of using open-source datasets for training LLMs?

a. Authenticity and credibility of real-world data

b. Guaranteed high accuracy in model predictions

c. Ability to acquire timely data and share data freely

d. Elimination of bias in training data

e. LLMs no longer require fine-tuning

Domanda 8 *(3 points (no penalty for wrong answers))*

Consider a scenario where you have:

- A natural language description of a function's requirements (REQS)
- A set of test cases (TESTS) specifying expected inputs and outputs

You need to design an agentic architecture using LLAMA-based agents to:

- Generate an initial implementation of the function based on REQS
- Refine the code iteratively until all test cases pass

Describe the architecture of LLAMA agents that perform these steps. For each agent:

- Define its role in the pipeline
- Specify its prompt, including:
  - System message: Defines the agent's behavior
  - Context: Information given to the agent from previous steps
  - Possible inputs: Data received from prior agents
  - Expected output: The structured result the agent should return

Notice: It is not necessary to write Python calls to transformers, code for output parsing or actual code for test execution and parsing. Focus on the agentic workflow and prompt engineering techniques.

Domanda 9 *(2 points (no penalty for wrong answers))*

Consider the following cosine similarity table. Which of the following threshold is more dependable in terms of TPR and FPR? $T = 0.70, 0.80, 0.90$

	Exp1	Exp2	Exp3	Exp4
Act1	0,71	0,65	0,62	0,30
Act2	0,71	0,74	0,30	0,20
Act3	0,54	0,05	0,60	0,55
Act4	0,80	0,40	0,15	0,92

(Note: "Exp" likely refers to Expected/True class or Experiment, and "Act" to Actual/Predicted, or vice-versa. Based on typical confusion matrix problems, assume diagonals are True Positives).

Domanda 10 *(1 point (-25% penalty for each wrong answer))*

Which solutions can help address the challenge of variation in extracted requirements?

a. Automatically rejecting extracted requirements with different phrasing

b. Ignoring minor variations and focusing only on exact matches

c. Using predefined synonym lists to map similar terms

d. Using word embeddings to detect semantically similar words

e. Manually reviewing every extracted requirement

Domanda 11 *(1.5 points (-25% penalty for a wrong answer))*

The following reward can be used when fine-tuning a model on human feedback.

$$R(x, y) = r_\theta(x, y) - \beta \log \frac{\pi_{new}(y|x)}{\pi_{old}(y|x)}$$

Where r_θ is the output of a Reward Model, π_{new} and π_{old} are the new and the original policies, respectively (you can see them as the probabilities assigned by the new and original models).

What is the purpose of the term $\log \frac{\pi_{new}(y|x)}{\pi_{old}(y|x)}$?

a. it increases the entropy of the output (i.e., it helps the model produce more "creative" answers)

b. it improves the model's capability of following instructions

c. it prevents the model from collapsing into a model that always predicts a single token

d. it helps make the model more deterministic

e. none of the other answers is correct

f. it prevents the new model from producing outputs that are too dissimilar from the original one

g. it rewards the model when it generates an answer that humans will "like"

Domanda 12 *(1 point (-25% penalty for each wrong answer))*

What is "priming" in the context of prompt engineering?

a. Using reinforcement learning to fine-tune the model
b. Pre-training the model with additional data
c. Providing initial input to guide the model's response
d. Adding noise to the input to test model robustness
e. Resetting the model before every new prompt

Soluzioni

Domanda 1:

- The code produces an error because `AutoModel` (base BERT) does not support `.generate()`. BERT is an encoder-only model, not built to produce new tokens autoregressively.

Domanda 2:

- (e) To evaluate past decisions and improve future performance.

Domanda 3:

- (g) Encoder self-attention only uses information from the input sequence, while encoder-decoder cross-attention allows the decoder to attend to the encoder's output.

Domanda 4:

- 1st merge: a , a
- 2nd merge: a , c
- 3rd merge: ac , e

Domanda 5:

- (g) It computes token-level similarity by aligning contextualized embeddings from a pre-trained transformer model and aggregating scores.

Domanda 6:

- quantized(x_1) = -127
- quantized(x_2) = 102
- quantized(x_3) = -25
- quantized(x_4) = 0

Domanda 7:

- Authenticity and credibility of real-world data
- Ability to acquire timely data and share data freely

Domanda 8: *(Summary of architecture)*

- **Agent 1: Coder/Implementer.** Role: Generate initial code. System: Expert Python developer. Input: REQS. Output: Python code block.
- **Agent 2: Tester/Refiner.** Role: Run tests and refine code. System: Expert debugger. Input: Code + Test Results. Output: Refined code.
- *Workflow:* Iterative loop where Agent 2 receives feedback (failed tests) and improves the code until all tests pass.

Domanda 9:

- The answer is $T = 0.7$.
- *Reasoning (from student attempt):* For $T = 0.7$, TP=3, FP=1, TN=10, FN=2. TPR=0.6, FPR=0.09.

Domanda 10:

- Using predefined synonym lists to map similar terms.
- Using word embeddings to detect semantically similar words.

Domanda 11:

- (f) it prevents the new model from producing outputs that are too dissimilar from the original one.

Domanda 12:

- (c) Providing initial input to guide the model’s response.